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MILITARY SPECIFICATION

RADAR BOMBING EQUIPMENT, AN/APA-16

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope: This specification establishes the requirements for the manufacture and testing of the AN/APA-16 Aircraft Radar Equipment, which is a bombing attachment for various airborne radars, for use on military aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on the date of invitation for bids, forms a part of this specification.

SPECIFICATIONS:

Military

MIL-D-5028(ASG)	Drawings and Data Lists; Preparation of Manufactures' (For Production Aircraft, Guided Missiles, Engines, Accessories and other Auxiliary Equipment)
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-E-17555(Ships)	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing and Marking of
MIL-R-18983(Aer)	Radar Bombing Equipment AN/APA-16; Installation Instructions for
MIL-R-18985(Aer)	Radar Bombing Equipment AN/APA-16; Bench, Preflight, Flight Test Installation for

STANDARDS:

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS:

AND 10398	Metals - Definition of Dissimilar
PT No. ST 1516	Manufacturer Drawings' List, Numerical, Emerson Radio and Phonograph Corporation
PT No. ST 1517	Manufacturer Drawings' List, Alpha-Numerical, Emerson Radio and Phonograph Corporation

PUBLICATIONS:

Air Force-Navy Aeronautical Bulletins No. 143	Specifications and Standards; use of

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(When requesting specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention Technical Records Division.)

3. REQUIREMENTS

3.1 Components: The complete aircraft radar equipment shall consist of a control unit, a range marker unit, a capacitor unit, mountings for these units, and a high voltage cable.

3.2 Materials: Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Metals: Metals shall be of the corrosion-resisting type or suitably treated to resist corrosion in fuels, salt spray, or atmospheric conditions to which the equipment shall be subjected when in storage or during normal service use.

3.2.1.1 Dissimilar Metals: Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Drawing AND10398.

3.2.2 Fungus-Proof Materials: Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent acceptable to the procuring activity.

3.2.3 Selection of Materials: Specifications and standards for all materials, parts, and government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143.

3.2.3.1 Standard Parts: Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2.3.2 Critical and Strategic Materials: Materials shall be chosen on the basis of suitability and availability in this country in quantity and from consideration of the additional restrictions created during a national emergency. Critical and strategic materials shall be used only to the extent considered mandatory for satisfactory operation. The contemplated use of these materials shall be brought to the attention of the procuring service with appropriate justification therefor and shall be subject to approval of the procuring service.

3.2.3.3 Weight Reduction: The contractor shall make every effort to reduce the over-all weight without impairing the equipment's function, operation or durability. All changes that are accomplished to require weight reduction shall be subject to written approval by the Procuring Service prior to incorporation.

3.2.3.4 Conformance to Drawings: No departure from, revision or change to the drawings specified herein shall be permitted without prior written approval by the Procuring Service. In the event changes are approved, (see paragraph 6.1) the contractor shall furnish the Procuring Service with reproducible drawings covering the change. Conformance to drawings will not relieve the contractor of the requirements of paragraphs 3.2.3.5 and 3.8.

3.2.3.5 Manufacturing and Finish: Manufacturing processes and finishes shall be as shown on the applicable drawings unless otherwise approved. (See paragraph 6.1). The responsibility shall be assumed by the contractor for any malfunction or failure of the component parts or the assembled equipment to pass the required tests due to unsatisfactory fits, clearances, tolerances, internal stresses or finishes.

3.3 Design: Model AN/APA-16 aircraft radar equipment is a low altitude radar bombsight attachment designed for use with search radars. It is intended primarily for use in bombing surface vessels from aircraft, but can be used against any target within radar range.

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3.3.1 Control Unit: Control Unit C-275A/APA-16 shall contain all of the external controls necessary to the operation and calibration of the equipment. It shall contain circuits which form an electrical computer which will be used to determine the bomb release range, and other circuits which supply D-C biasing voltages to control the action of the Range Marker circuits. The value of this bias voltage is determined by data set up in the Control Unit.

3.3.2 Range Marker Unit: Range Marker Unit SN-12B/APA-16 participates in all the electronic circuits of the equipment. It also contains the power supply used to furnish plate and filament voltages to the Range Marker circuits and D-C voltages to the Control Unit.

3.3.3 Capacitor Unit: Capacitor Unit MX-139/APA-16 contains a specially constructed one-microfarad capacitor which functions as an integral part of the rate-sweep generator circuit in the Range Marker Unit. This capacitor is housed separately to protect it from excessive heat.

3.3.4 Mounting Bases: The three above-mentioned units are mounted to the aircraft structure by Mounting Base MT-208/APA-16 for the Range Marker Unit, Mounting Plate MT-209/APA-16 for the Control Unit, and Mounting Plate MT-210/APA-16 for the Capacitor Unit.

3.3.5 H V Cable: The high voltage cable is used to connect Capacitor Unit MX-139/APA-16 to the Range Marker Unit. This is a complete cable-and-plug assembly consisting of a 5 foot length of RG-8/U coaxial cable with a UG-180A/U connector at each end.

3.3.6 Adapter Plug: Adapter Plug U-47/APA-16 is used as a dummy plug on the Control Unit when the Adapter Unit TD-6/APA-16 is not in use. (Adapter Unit TD-6/APA-16 is not supplied by the AN/APA-16 contractor.)

3.4 Interchangeability: All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Detailed Requirements and Properties Electrical: The AN/APA-16 requires a 115 volt, 400 cycle power source, and must be supplied with a trigger pulse in time coincidence with the radar trigger pulse. The equipment must be capable of operation at pulse repetition rates up to 2,000 cycles per second. Two pulses must be produced by the equipment; a range marker pulse and a release pulse. Each of these pulses should be 0.5 microseconds wide at the base and at least 15 volts peak in magnitude.

3.6 Performance: The equipment shall satisfy the performance requirements (conform to the test limits) of all tests in Section 4.

3.7 Nameplate: A nameplate, conforming to the requirements of Specification MIL-M-7911, shall be securely attached to each component.

3.8 Workmanship: All details of workmanship shall be in accordance with high-grade armament manufacturing practice for the quantity production of this type equipment. All parts shall be free from excessive marks, grooving, indentations, cracks, flaws, and any other injurious defects which may impair the serviceability of the assembled equipment. All workmanship shall be of a quality satisfactory to the Procuring Service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests: The inspection and testing of Aircraft Radar Equipment AN/APA-16 shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the item after the award of the contract, to determine that the production meets the requirements of this specification.

(b) Inspection tests: Inspection tests are those tests accomplished on Aircraft Radar Equipment AN/APA-16 manufactured and submitted for acceptance under contract.

4.2 Preproduction Tests:

4.2 Sampling Instructions: The first unit of the aircraft radar equipment under each contract shall be preproduction test samples and shall be delivered for inspection and testing as

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directed, together with a certification by the cognizant inspector that the materials used in their manufacture are in accordance with pertinent specifications.

4.2.2 Tests: The preproduction samples shall be checked for strict conformance with the pertinent drawings and specifications, and shall be subjected to any or all tests of this specification, as described under "Test Methods."

4.2.3 Correction of Defects: Any defects discovered during the inspection and testing of the preproduction samples shall be corrected by the contractor. Such corrective measures as may be found necessary shall likewise be incorporated in all subsequent equipment offered for acceptance under the contract.

4.3 Materials Inspection:

4.3.1 Commercial Materials: When materials are procured under commercially recognized non-Government specifications, such as by SAE numbers, a certificate showing compliance with such specifications shall be furnished by the supplier of such material. Such certificate shall be in a form acceptable to the Government Inspector.

4.3.1.2 Approval of Materials in Process: Inspection and approval of material in process during the course of manufacture shall not be construed as insuring the acceptance or approval of the finished product.

4.3.2 Inspection Tests: Inspection tests shall consist of Individual tests and Sampling tests. The equipment, together with all the materials used in the manufacture thereof, shall be subjected to inspection at the contractor's plant, or elsewhere, by authorized Government inspectors.

4.3.2.1 Individual Tests: Each Aircraft Radar Equipment AN/APA-16 shall be subjected to the following tests. These tests are Release Range and associated calibrations for the equipment, and shall be performed by the contractor.

4.3.2.1.1 Test Conditions for Individual Tests: This series of tests shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and a temperature of approximately 25° C. (77° F.). Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance for change in instrument reading shall be made for the difference from specified conditions. The equipment shall be tested with 115 volt, 400 cycle alternating current. The voltage and frequency tolerance shall be ±2 percent.

4.3.2.1.2 Equipment Starting Time:

(a) Set all the controls for operation and observe the time from the instant the equipment is turned on until the range mark appears on the oscilloscope screen.

(b) The maximum time recorded should not exceed 30 seconds.

4.3.2.1.3 Equipment Stability at Start:

(a) Observe that there is no instability in the range mark on the oscilloscope as the power to the equipment is turned on and off at one half minute intervals.

(b) The pattern on the oscilloscope should remain stable. With S-201 in position 1, the pattern should be at least 20 volts; when in position 2, the pattern should be at least 15 volts.

4.3.2.1.4 Reference Range Zero: Set up the controls in the Control Unit as follows:

Control	Position
POSITION	Nearly fully clockwise
BOMB SCALE ADJ	Midway of its range
CALIBRATE SWITCH	REF RANGE ZERO
BOMB CORRECTION	Zero feet
RANGE ZERO	Fully counterclockwise
SEARCH-TRACK	SEARCH

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4.3.2.1.4.1 Switch to TRACK and turn the RANGE ZERO control slowly clockwise until the dial lights go out.

4.3.2.1.4.2 Check this setting several times by turning the BOMB CORRECTION dial toward -100, switching to TRACK, and then advancing the BOMB CORRECTION dial slowly until the lights go out again.

4.3.2.1.4.3 The BOMB CORRECTION dial should read 0 ±5 feet.

4.3.2.1.5 Range Zero - Set up the controls in the Control Unit as follows:

Control	Position
CALIBRATE SWITCH	RATE OR RLS (RELEASE) RANGE ZERO
BOMB CORRECTION	∕ 50 feet

4.3.2.1.5.1 Switch to TRACK and turn the PHASE ADJ control of the calibrator slowly in a counterclockwise direction until the dial lights go out.

4.3.2.1.5.2 Check this setting several times, each time turning the BOMB CORRECTION dial toward -100, switching to TRACK, and then advancing the BOMB CORRECTION dial slowly toward ∕ 50 until the dial lights again go out.

4.3.2.1.5.3 The dial reading should be ∕ 50 ±5 feet.

4.3.2.1.6 Calibrating the Bomb Correction Scale: As a preliminary step in this test, it is necessary to adjust the computer in the Control Unit so that, for the settings of the CLOSING RATE and ALTITUDE dials specified, the release range pulse occurs at the correct distance from range zero. Set up the controls in the Control Unit as follows:

Control	Position
CLOSING RATE	325 knots
BOMB CORRECTION	∕ 50 feet
ALTITUDE	481 feet
CALIBRATE SWITCH	RELEASE RANGE ADJ
RLS RANGE ADJ	Maximum counterclockwise

4.3.2.1.6.1 Switch to TRACK and turn the RLS RANGE ADJ control slowly in a clockwise direction until the dial lights go out. Check the setting with the BOMB CORRECTION dial as in previous tests. Lights should go out at ∕ 50 ±5 feet.

4.3.2.1.6.2 Retard the BOMB CORRECTION Dial toward -100, turn the CALIBRATE SWITCH to RATE OR RLS RANGE ZERO, and check range zero with the BOMB CORRECTION dial. The reading should be ∕ 50 ±10 Feet.

4.3.2.1.6.3 Return the CALIBRATE SWITCH to RELEASE RANGE ADJ. Check the BOMB CORRECTION reading as before, and readjust the RLS RANGE ADJ control. Continue two previous steps until the dial lights go out on ∕ 50 ± 10 feet on both positions of the CALIBRATE SWITCH.

4.3.2.1.6.4 Move the ALTITUDE dial to read 348 feet. Switch to TRACK, and advance the BOMB CORRECTION dial slowly toward ∕ 500; the dial lights should go out when ∕ 500 is reached.

4.3.2.1.6.5 Repeat checks of the settings in four previous steps, adjusting respective controls if necessary since the RANGE ZERO and BOMB SCALE ADJ settings are reached by a process of successive approximations. When properly adjusted, the spread between readings on the BOMB CORRECTION dial for ALTITUDE dial settings of 481 and 348 feet should be 450 ±20 feet.

4.3.2.1.7 Check of Computer Circuits: The procedure is to set the RLS RANGE ADJ control with the CLOSING RATE and ALTITUDE dials set for a combination of values which will give a 1,500 foot or 3,000 foot release point, choosing these settings from the sample Table 1, to correspond as nearly as possible to the average altitude and closing rates used in bombing runs. When this adjustment is completed, check the other combinations of altitude and closing rate given in the sample table taking,

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for each, a reading on the BOMB CORRECTION dial, first, with the CALIBRATE SWITCH on RATE OR RLS RANGE ZERO, and second, with it on RELEASE RANGE ADJ. The resulting reading of the BOMB CORRECTION dial being entered in the appropriate column (4) or (5) of such a table. The method of making these checks is given in the following steps.

4.3.2.1.7.1 Set the CALIBRATE SWITCH on RATE OR RLS RANGE ZERO, the BOMB CORRECTION dial at -100, switch to TRACK, and advance the BOMB CORRECTION dial slowly clockwise until the dial lights are extinguished. The reading of the BOMB CORRECTION dial is entered in column (4) of the table.

4.3.2.1.7.2 Set the CALIBRATE SWITCH on RELEASE RANGE ADJ, the BOMB CORRECTION dial at -100, switch to TRACK and advance the BOMB CORRECTION dial slowly clockwise until the dial lights are extinguished. The reading of the BOMB CORRECTION dial is entered in column (5) of the sample table.

4.3.2.1.7.3 Make the initial setting with the selected CLOSING RATE and ALTITUDE dial readings. For example: 175 knots and 415 feet for 1,500 feet release point; 325 knots and 481 feet for 3,000 feet release point. The spread between the BOMB CORRECTION dial readings should be within the tolerance of ± 10 feet.

4.3.2.1.7.4 Without changing the RANGE ZERO or RLS RANGE ADJ settings, make checks on other combinations of closing rate and altitude shown in the sample table.

4.3.2.1.7.5 Find the algebraic difference between the readings in columns (4) and (5), and enter the result in column (6). See examples in table. The difference for any given combination of CLOSING RATE and ALTITUDE dial settings should not exceed 50 feet, and the algebraic difference between the maximum and minimum error should not exceed 60.

4.3.2.1.8 Rate Zero Check: Set up the controls of the Control Unit and test equipment as follows:

Control	Location	Position
POSITION	Control Unit	Nearly fully clockwise
ALTITUDE	Control Unit	50 feet
BOMB CORRECTION	Control Unit	-100 feet
CALIBRATE SWITCH	Control Unit	OPERATE
CLOSING RATE	Control Unit	300 knots
RANGE MARK	Control Unit	Adjust as required
MARKER POLARITY	Calibrator	- (Minus)

4.3.2.1.8.1 Adjust the test oscilloscope to obtain clear-cut images. If Oscilloscope CW-60ACB is used, the image amplitude should be adjusted to one-half inch or less to prevent overloading.

4.3.2.1.8.2 While observing the test oscilloscope screen, turn the SEARCH-TRACK switch on the Control Unit to TRACK.

4.3.2.1.8.3 The range and release markers should appear on the sweep trace as narrow vertical pulses. The calibrating pulses will appear as equispaced bright spots along the sweep trace.

4.3.2.1.8.4 Use the POSITION control to set the tracking edge of the range mark image within about 1/16 inch of the nearest calibrating marker.

4.3.2.1.8.5 There should be no relative motion between the range mark and the calibrating marker for a period of one minute, by actual observation.

4.3.2.1.9 Rate Calibration: Set the controls on the Control Unit and test equipment as follows:

Control	Location	Position
POSITION	Control Unit	Nearly fully clockwise
ALTITUDE	Control Unit	50 feet

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TABLE 1

SAMPLE TABLE FOR CHECK OF COMPUTER

The values in Columns (4), (5) and (6) are for illustrative purposes only.

Dial Settings			BOMB CORRECTION Dial Reading with CALIBRATE SWITCH on		Algebraic Difference Column (4) minus Column (5) Error
CLOSING RATE (knots)	ALTITUDE (feet)	RELEASE RANGE (feet)	RATE OR RLS RANGE ZERO	RLS RANGE ADJ	
(1)	(2)	(3)	(4)	(5)	(6)
325	481	3,000			
350	415	3,000	+60	+30	+30
375	361	3,000			
400	318	3,000			
400	79.4	1,500			
375	90.3	1,500			
350	104	1,500	+50	+10	+40
325	120	1,500			
300	141	1,500			
275	168	1,500			
250	203	1,500	+50	+60	-10
225	251	1,500			
200	318	1,500			
175	415	1,500			

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Control (cont.)	Location (cont.)	Position (cont.)
BOMB CORRECTION	Control Unit	-100 feet
CALIBRATE SWITCH	Control Unit	OPERATE
CLOSING RATE	Control Unit	As specified in text below
RANGE MARK	Control Unit	Released
SWEEP SPEED	TS-34/AP	MED
SWEEP TIME	CW-60ACB	NORMAL
MARKER POLARITY	Calibrator	- (minus)

4.3.2.1.9.1 Set the CLOSING RATE dial to 148 knots and, while observing the oscilloscope screen, switch to TRACK. The range mark should move to the left along the sweep trace. Beginning with the third marker to appear, carefully measure the time required to pass five additional markers. It should require 30 seconds. If it does not, correct the RATE ADJ control, turning it clockwise if the elapsed time is greater, and counterclockwise if less than it should be. Check the time required to uncover or pass other groups of five timing markers, resetting the SEARCH-TRACK if necessary. Make the final adjustment accurately on a group of five markers near the bottom of the screen, after switching to the shortest sweep. Do not use the last two timing markers to appear.

4.3.2.1.9.2 Upon completion of this adjustment, recheck. Set the range mark to maximum range with the POSITION control; switch to TRACK, and, beginning with the second marker to appear, time consecutive groups of five markers. Enter the results in a table similar to Table II. Repeat the procedure at the other rates shown in the table, entering the results in the proper line and column.

TABLE II

SAMPLE TABLE FOR RATE CALIBRATION

CLOSING RATE (knots)	Time required to Uncover Five Timing Markers			
	Theoretical	1st Group of 5 Markers	2nd Group of 5 Markers	3rd Group of 5 Markers
(1)	(2)	(3)	(4)	(5)
74	60			
111	40			
148	30			
222	20			
296	15			
369	12			

4.3.2.1.9.3 Along a line in this table corresponding to any one of the closing rates, the largest reading should not differ from the smallest by more than 2%. Values in the last column should not depart from the theoretical value by more than ±1% for rates up to 300 knots. At rates above 300 knots ±2% is acceptable. If these tolerances cannot be met, check the calibrator, since it has large pulse-rate tolerances.

4.3.2.1.10 Jitter Readings: Jitter is determined by comparing time necessary to uncover single markers.

4.3.2.1.10.1 Set the CLOSING RATE at 200 knots; switch to TRACK.

4.3.2.1.10.2 Make eight readings.

4.3.2.1.10.3 The 2nd, 5th and 8th readings should be between 4.22 and 4.66 seconds. The maximum difference between any of these readings should not exceed 0.25 seconds.

4.3.2.1.11 Release Relay Operating Time: This test measures the time delay between operation of relays K-201 and K-202.

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4.3.2.1.11.1 Switch to TRACK.

4.3.2.1.11.2 Take four readings and then return switch S-3 to OFF.

4.3.2.1.11.3 The measurements should fall between 2.2 and 3.8 seconds.

4.3.2.1.12 Modulation Test

4.3.2.1.12.1 With the REPETITION FREQUENCY of the RANGE CALIBRATOR set at 800 cycles, slowly vary the power supply frequency between 750 and 850 cycles. Note if there is any evidence of instability in the range mark, particularly at its leading and lagging edges.

4.3.2.1.12.2 If varying the frequency causes variation in the A-C voltages, note the amount of variation. Voltage variations will cause a variation in the leading or release edge of the pulse of about 12 feet per volt and this should not be confused with variation due to frequency.

4.3.2.1.12.3 Repeat the test at a pulse repetition rate of 400 cycles, varying the power supply frequency between 390 and 450 cycles. (Do not let the frequency get much below 400 cycles as the transformers in the equipment are not designed for lower frequencies.)

4.3.2.1.12.4 Repeat the test at a pulse repetition rate of 1600 cycles, varying power supply frequency between 1550 and 1650 cycles.

4.3.2.1.13 Reference Range Zero Drift: Check the REFERENCE RANGE ZERO and see if reading is within ±10 feet of last reset reading. If variation is more than 10 feet, check to see that drift does not exceed 10 feet within a 10 minute period.

4.3.2.2 Sampling Tests: Sampling tests shall consist of all individual tests as listed in Paragraph 4.3.2.1 and its sub-paragraphs with the addition of the D.C. Voltages test in Paragraph 4.3.2.2.2, but under the conditions listed in Paragraph 4.3.2.2.1. The Sampling tests shall be applied on a 2% of production basis, selected at random by the Government Inspector.

4.3.2.2.1 Test Conditions for Sampling Tests: The various Sampling tests shall be made under the applicable conditions in the list below. Reference should be made to the Sampling Tests Data Sheet (attached as a part of this specification) as a summary of procedural information and as an example of the form recommended for recording Sampling tests' results.

4.3.2.2.1.1 Normal Conditions Test: Install the equipment in the temperature chamber and check that all controls operate smoothly. With the chamber door open turn on set and make the specified electrical tests as required on the Data Sheet. Turn the equipment to OFF.

4.3.2.2.1.2 Low Temperature Test: Close the chamber door and stabilize the temperature at -55°C. for 1 hour (equipment at OFF). Raise the chamber temperature to -40° C. for 1 hour; then lower the pressure in the chamber to the equivalent of 20,000 feet. Turn the equipment to ON, make the specified measurements (see Data Sheet), and continue operation of equipment until at least one hour has elapsed from start of electrical tests. Turn the equipment to OFF.

4.3.2.2.1.3 Frosting Test: With the equipment power at OFF change the chamber conditions to ambient altitude and -10° C. temperature. Hold for one hour. Open the chamber door and allow frost to form on the equipment. Do not allow enough time to permit equipment to dry out. Leave door open long enough to melt the frost. Turn the equipment to ON and make the required tests.

4.3.2.2.1.4 Normal Conditions Test: Stabilize the equipment at room temperature with the chamber door open, and make the required measurements.

4.3.2.2.1.5 Low Voltage (High Temperature) Tests: With equipment power at OFF, raise the chamber temperature to +85° C. (Remove the capacitor at or below +70° C.) and hold the equipment at +85°C. for one hour; then lower to 55° C. (replace the Capacitor at or below +70° C.) and hold for one hour. Turn on equipment at 115 volts and 800 cycles. Take the specified measurements as per Data Sheet. Reduce the voltage to 109.25 volts, operate for 5 minutes, and take the required readings. Continue operation throughout this test at 109.25 volts. Reduce the frequency to 400 cycles and make the electrical tests.

4.3.2.2.1.6 High Voltage (High Temperature) Test: This test should be made immediately following the low voltage test and without turning off the power or changing the chamber conditions (+55° C.). Operate at 115 volts for 5 minutes (800 cycles); then take readings. Change the voltage to 120.75 and maintain this voltage for the remainder of the test. Operate for 5 minutes at 800 cycles and

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take readings. Change to 400 cycles, take the specified readings, and make the 2 hour run at various frequencies as specified for the low voltage test. Take readings at 1800 cycles just before the end of the test.

4.3.2.2.1.7 Normal Voltage (High Temperature) Test: Following completion of the high voltage test, reduce the voltage to 115 and frequency to 800 cycles. Operate for 5 minutes under these conditions; then make the specified measurements.

4.3.2.2.1.8 Humidity Tests: With the equipment non-operative (OFF), stabilize the equipment at +55°C. Adjust the humidity to 95% and hold for 6 hours. Maintain the humidity as near 95% as practicable while changing the temperature to +10°C. for 6 hours, then at +55°C. for 6 hours, and finally +10°C. for 6 hours. At the end of the 24 hour run, turn the equipment to ON and make the required measurements. Dry out the equipment.

4.3.2.2.1.9 Vibration Test: Mount the equipment on the vibration machine with the major axes parallel. Make the specified electrical measurements before vibration. Vibrate with 0.06 inch excursion in each of the lateral, longitudinal and vertical planes. Vibrate for one half hour in each plane with frequency changing from 10 to 60 cycles per second, plus 5 minutes at each resonant frequency. Observe the range mark on the oscilloscope during vibration to insure that instability is not present. After vibration make another series of electrical measurements.

4.3.2.2.1.10 Visual Inspection: All units shall be inspected to insure conformity with the requirements or the applicable drawings insofar as dimensions, location of mounting holes, finishes, etc.

4.3.2.2.2 D. C. Voltages Test: Read the D. C. voltages in the Range Marker Unit. The voltages are nominally +150, +250, and -150 volts.

4.3.2.3 Rejection and Retest: Items which have been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 All components of the Radar Bombing Equipment, AN/APA-16, shall be preserved, packaged, packed and marked in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use:

6.2 Adherence to Drawings and Specifications - Conflict Elimination: Insofar as the contractor is required by provisions of this specification to adhere rigidly to the drawings listed in Section 2, which may make it impossible in some instances also to comply with the performance and test requirements specified herein, then, in that event, the contractor should promptly notify the procuring activity of such fact, and appropriate changes will be made in the applicable drawings, parts, or performance requirements as authorized by the procuring activity. An equitable adjustment, in accordance with applicable contract provisions, will be made in the price of the articles to be delivered to reflect any change so authorized by the procuring activity.

6.3 Ordering Data: Each bid submitted in connection with the procurement of the equipment shall be accompanied by a statement showing whether or not any exceptions are being taken to the requirements shown in the drawings or in this specification. If exceptions are taken, they shall be specifically described, and the alternative thereto indicated in the statement.

PATENT NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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HIGH VOLTAGE TEST HIGH TEMPERATURE

Temperature_____ Date_____
Pressure_____ Tester_____
Humidity_____ Gov't Insp._____

Start of Test_____

A.C. VOLTAGE 115 VOLTS, FREQUENCY 800 CYCLES

1. Reference Range Zero_____

2. D. C. voltages +150_____ +250_____ -150_____

3. Jitter readings

1st rdg. _____ 4th rdg. _____ 7th rdg. _____
2nd rdg. _____ 5th rdg. _____ 8th rdg. _____
3rd rdg. _____ 6th rdg. _____ (Limit 2nd, 5th, and 8th rdgs. bet. 4.22 and 4.66 sec. and max. diff. bet. these rdgs. of 0.25 sec.)

4. Release relay operating time (2.2 to 3.8 sec.)

1st rdg._____sec. 3rd rdg._____sec.
2nd rdg._____sec. 4th rdg._____sec.

A.C. VOLTAGE 120.75 VOLTS, FREQUENCY 800 CYCLES

5. Reference Range Zero_____

6. D. C. voltages +150_____ +250_____ -150_____

7. Jitter Readings

1st rdg. _____ 4th rdg. _____ 7th rdg. _____
2nd rdg. _____ 5th rdg. _____ 8th rdg. _____
3rd rdg. _____ 6th rdg. _____ (Limit 2nd, 5th, and 8th rdgs. bet. 4.22 and 4.66 sec. and max. diff. bet. these rdgs. of 0.25 sec.)

8. Release relay operating time (2.2 to 3.8 sec.)

1st rdg._____sec. 3rd rdg._____sec.
2nd rdg._____sec. 4th rdg._____sec.

A.C. VOLTAGE 120.75 VOLTS, FREQUENCY 400 CYCLES

9. Reference Range Zero_____

10. D. C. voltages +150_____ +250_____ -150_____

11. Jitter readings:

1st rdg._____ 4th rdg._____ 7th rdg._____
2nd rdg._____ 5th rdg._____ 8th rdg._____
3rd rdg._____ 6th rdg._____
(Limit 2nd, 5th, and 8th rdgs. bet. 4.22 and 4.66 sec. and max. diff. bet. these rdgs. of 0.25 second)

12. Release relay operating time (2.2 to 3.8 sec.)

1st rdg._____sec. 3rd rdg._____ sec.
2nd rdg._____sec. 4th rdg._____ sec.

A.C. VOLTAGE 120.75 VOLTS, FREQUENCY 1800 CYCLES

13. Reference Range Zero_____

14. D. C. Voltages +150_____ +250_____ -150_____

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HIGH VOLTAGE TEST HIGH TEMPERATURE (cont.)

15. Jitter readings

1st rdg._____	4th rdg. _____	7th rdg._____
2nd rdg._____	5th rdg. _____	8th rdg._____
3rd rdg._____	6th rdg. _____	

16. Release relay operating time (2.2 to 3.8 sec.)

1st rdg. _____sec.	3rd rdg._____sec.
2nd rdg. _____sec.	4th rdg._____sec.

LIMITS FOR ALL JITTER READINGS:
(Limit 2nd, 5th, and 8th rdgs. between 4.22 and 4.66 seconds and max. diff. between these rdgs. of 0.25 seconds)

NOTES:

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HIGH TEMPERATURE TEST NORMAL VOLTAGE CHECK

Temperature_____
Pressure_____
Humidity_____
Power Freq._____
A-C Voltage_____

Date_____
Tester_____
Gov't Insp._____

Start of Test_____

1. Reference range zero before reset_____(Limit ± 10' in 10 min.)

2. D. C. voltages +150_____ +250_____ -150_____

3. Ref. range zero final reset rdg._____(0 ± 5)

4. Release range zero rdg. _____(50 ± 5)

5. Release range errors

Closing Rate Knots (1)	Alt. Feet (2)	Bomb Correction Rdgs. Calibrate Switch on Rate or Rls Range Zero (3)	Release Range Adj. (4)	Alg. Diff. (3)-(4) Error (5)	Limit
325	481	_____	_____	_____	(50 max.)
400	79.4	_____	_____	_____	"
175	415	_____	_____	_____	"

Col (5) limit, 60 max. algebraic difference between rdgs.

6. Rate and rate linearity - before adjusting rate zero

	148 Knots
1st rdg.	_____
2nd rdg.	_____
3rd rdg.	_____
Average	_____

Limit 23.76 - 24.24

7. Rate Zero
Direction of Drift_____
Was adjustment necessary?_____

Left (toward range zero) or right (away from range zero)

8. Rate and rate linearity - after adjusting rate zero

	148 Knots
1st rdg.	_____
2nd rdg.	_____
3rd rdg.	_____
Average	_____

Difference between larger and smaller shall be no greater than 2% of smaller.

9. Jitter readings

1st reading_____	4th reading _____	7th rdg._____
2nd reading_____	5th rdg. _____	8th rdg._____
3rd rdg._____	6th rdg. _____	

(Limit 2nd, 5th, and 8th rdgs. between 4.22 and 4.66 seconds and max. diff. between these rdgs. of 0.25 seconds.)

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HIGH TEMPERATURE TEST NORMAL VOLTAGE CHECK (cont.)

10. Release relay operating time (2.2 to 3.8 sec.)

1st rdg._____sec. 3rd rdg._____sec.
2nd rdg._____sec. 4th rdg._____sec.

11. Was instability present in operation of position control?

_____

NOTES:

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HUMIDITY TEST AFTER HUMIDITY RUN

Temperature_____
Pressure_____
Humidity_____
Power Freq._____
A-C Voltage_____

Date_____
Tester_____
Gov't Insp._____

Start of Test_____

1. Elapsed time until range marks appears_____sec. (30 sec. max.)

2. Was instability noted as power was turned on and off 5 times at 1/2 minute intervals?

3. Reference range zero
   Initial rdg. _____ 3rd rdg._____
   1st reset rdg._____ 4th rdg._____
   2nd rdg._____ 5th rdg._____
   Limit + or - 10' in 10 min.

4. D. C. Voltages +150_____ +250_____ -150_____

5. Ref. range zero final reset rdg._____(0 ± 5)

6. Release range zero rdg._____(50 ± 5)

7. Release range errors

Closing Rate Knots (1)	Alt. Feet (2)	Bomb Correction Rdgs. Calibrate Switch On: Rate or Rls Range Zero (3)	Release Range Adj. (4)	Alg. Diff. (3)-(4) Error (5)	
325	481	_____	_____	_____	(50 Max.)
400	79.4	_____	_____	_____	"
175	415	_____	_____	_____	"

Col. (5) limit 60 max. algebraic difference between rdgs.

8. Rate and rate linearity - before adjusting rate zero

148 Knots

1st rdg. _____
2nd rdg. _____
3rd rdg. _____
Average _____
Limit 23.76 - 24.24

9. Rate Zero
   Direction of Drift_____
   Was adjustment necessary?_____

10. Rate and rate linearity - after adjusting rate zero

148 Knots

1st rdg. _____
2nd rdg. _____
3rd rdg. _____
Average _____

Difference between larger and smaller shall be no greater than 2% of smaller.

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HUMIDITY TEST AFTER HUMIDITY RUN (cont.)

11. Jitter readings

1st rdg._____	4th rdg._____	7th rdg._____
2nd rdg._____	5th rdg._____	8th rdg._____
3rd rdg._____	6th rdg._____	

(Limit 2nd, 5th and 8th rdgs. between 4.22 and 4.66 seconds and max. diff. between these rdgs. of 0.25 seconds.)

12. Release relay operating time (2.2 to 3.8 sec.)

1st rdg._____sec.	3rd rdg._____sec.
2nd rdg._____sec.	4th rdg._____sec.

NOTES:

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NORMAL ROOM CONDITIONS BEFORE VIBRATION

Temperature_____
Pressure_____
Humidity_____
Power Freq._____
A-C Voltage_____

Date_____
Tester_____
Gov't Insp._____

Start of Test_____

1. Elapsed time until range marks appears_____ sec. (30 sec. max.)

2. Was instability noted as power was turned on and off 5 times at 1/2 minute intervals?_____

3. Reference range zero
   Initial rdg._____
   1st reset rdg._____
   2nd rdg._____
   Limit + or - 10' in 10 min.
   3rd rdg. _____
   4th rdg. _____
   5th rdg. _____

4. D. C. Voltages +150_____ +250_____ -150_____

5. Ref. range zero final reset rdg. _____(0 ± 5)

6. Release range zero rdg._____(50 ± 5)

7. Release range errors

Closing Rate Knots (1)	Alt. Feet (2)	Bomb Correction Rdgs. Calibrate Switch On: Rate or Rls. Range Zero (3)	Release Range Adj. (4)	Alg. Diff. (3)-(4) Error (5)	
325	481	_____	_____	_____	(50 max.)
400	79.4	_____	_____	_____	"
175	415	_____	_____	_____	"

Col (5) max. 60 algebraic difference between rdgs.

8. Rate and rate linearity - before adjusting rate zero

	148 Knots
1st rdg.	_____
2nd rdg.	_____
3rd rdg.	_____
Average	_____

Limit 23.76 - 24.24

9. Rate Zero
   Direction of Drift_____
   Was adjustment necessary?_____

10. Rate and rate linearity - after adjusting rate zero

	148 Knots
1st rdg.	_____
2nd rdg.	_____
3rd rdg.	_____
Average	_____

Difference between larger and smaller shall be no greater than 2% of smaller.

11. Jitter readings

1st rdg._____ 4th rdg._____ 7th rdg._____
2nd rdg._____ 5th rdg._____ 8th rdg._____
3rd rdg._____ 6th rdg._____

(Limit 2nd, 5th, and 8th rdgs. between 4.22 and 4.66 seconds and max, diff. between these rdgs. of 0.25 seconds)

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NORMAL ROOM CONDITIONS BEFORE VIBRATION (cont.)

12. Release relay operating time (2.2 to 3.8 sec.)

1st rdg.	_____ sec.	3rd rdg.	_____ sec.
2nd rdg.	_____ sec.	4th rdg.	_____ sec.

NOTES:

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VIBRATION TESTS

Start of Test_____ Date_____
Tester_____
Gov't Insp. _____

1. Record troubles or defects noted during vibration tests, particularly resonant frequencies of each unit, axis on which they occur and maximum excursion at these frequencies.

(a) Range Marker Unit

Resonant Freq.	Axis	Max. Excursion
_____	VERTICAL	0.06"
_____	LATERAL	"
_____	LONGITUDINAL	"

(b) Control Unit

_____	VERTICAL	0.06"
_____	LATERAL	"
_____	LONGITUDINAL	"

(c) Capacitor Unit

_____	VERTICAL	0.06 "
_____	LATERAL	"
_____	LONGITUDINAL	"

(d) Other troubles or defects noted. NOTES:

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NORMAL ROOM CONDITIONS AFTER VIBRATION

Temperature_____
Pressure_____
Humidity_____
Power Freq._____
A-C Voltage_____

Date_____
Tester_____
Gov't Insp._____

Start of Test_____

1. Elapsed time until range marks appears_____sec. (30 sec. max.)

2. Was instability noted as power was turned on and off 5 times at 1/2 minute intervals?_____

3. Reference range zero
   Initial rdg._____
   1st reset rdg._____
   2nd rdg._____
   3rd rdg._____
   4th rdg._____
   5th rdg._____
   Limit + or - 10' in 10 minutes

4. D. C. Voltages +150_____ +250_____ -150_____

5. Ref. range zero final reset rdg._____ (0 ± 5)

6. Release range zero rdg._____(50 ± 5)

7. Release range erros

Closing Rate Knots (1)	Alt. Feet (2)	Bomb Correction Rdgs. Calibrate Switch On: Rate or Rls Range Zero (3)	Release Range Adj. (4)	Alg. Diff. (3)-(4) Error (5)	
325	481	_____	_____	_____	(50 max.)
400	79.4	_____	_____	_____	"
175	415	_____	_____	_____	"

Col (5) limit 60 max. algebraic difference between readings

8. Rate and rate linearity - before adjusting rate zero

	148 Knots
1st rdg.	_____
2nd rdg.	_____
3rd rdg.	_____
Average	_____

Limit 23.76 - 24.24

9. Rate Zero
   Direction of Drift_____
   Was adjustment necessary?_____

10. Rate and rate linearity - after adjusting rate zero

	148 Knots
1st rdg.	_____
2nd rdg.	_____
3rd rdg.	_____
Average	_____

Difference between larger and smaller shall be no greater than 2% of smaller.

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NORMAL ROOM CONDITIONS AFTER VIBRATION (cont.)

11. Jitter readings

1st rdg._____	4th rdg._____	7th rdg._____
2nd rdg._____	5th rdg._____	8th rdg._____
3 rd rdg._____	6th rdg._____	

(Limit 2nd, 5th, and 8th rdgs. between 4.22 and 4.66 seconds and max. diff. between these rdgs. - of 0.25 seconds.)

12. Release relay operating time (2.2 to 3.8 sec.)

1st rdg._____sec.	3rd rdg._____sec.
2nd rdg._____sec.	4th rdg._____sec.

NOTES:

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VISUAL INSPECTION

Date_____
Tester_____
Gov't Insp._____

Capacitor Unit-

Control Unit-

Range Unit -

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